

PROJECT REPORT ON

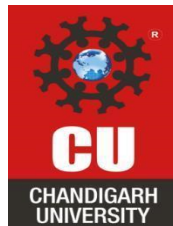
“ Daily Traffic Trend Analyzer ”

Submitted By:

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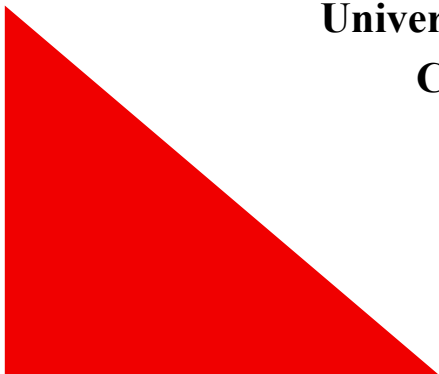
Under The Guidance of:

Dr. Sanjay Aggarwal



April, 2025

**University Institute of Computing
Chandigarh University,
Mohali, Punjab**





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I would like to express my sincere gratitude to all those who supported and guided me throughout the completion of this mini project, *"Daily Traffic Trend Analyser."*

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My gratitude goes to **Google Analytics** and **Blogger**, the tools that formed the technical backbone of this project. The availability of comprehensive documentation and user-friendly interfaces greatly assisted in the practical implementation and analysis.

Certified that this project report on *"Daily Traffic Trend Analyser"* is the Bonafide work of "Tushar Tomar" who carried out the project work under my supervision.

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HEAD OF THE DEPARTMENT

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Daily Traffic Trend Analyzer

1. Abstract

In today's competitive digital marketplace, data plays a vital role in shaping the success of e-commerce platforms. This mini project focuses on integrating Google Analytics with *Cara*, an online clothing e-commerce website, to monitor and analyze daily traffic trends and user behavior. The goal is to gain actionable insights that help improve user engagement, optimize website performance, and support data-driven decision-making.

Google Analytics was successfully linked to the *Cara* website using a unique tracking ID, allowing for the continuous collection of user data. Key performance indicators such as page views, sessions, bounce rates, session durations, and traffic sources were tracked to evaluate how users interact with the website. The data revealed crucial patterns such as peak visit hours, top-performing pages, and user flow through the site.

Traffic analysis showed that most users accessed the site via mobile devices, with organic search and social media being the primary sources of traffic. This insight highlights the importance of mobile optimization and targeted digital marketing strategies. Additionally, the checkout page had a relatively high bounce rate, indicating a potential area for UX improvement.

Visual representations of the data were created using charts and graphs to make insights easily understandable and actionable. Performance was further optimized using tools like Google PageSpeed Insights to ensure fast loading and a smooth browsing experience.

In conclusion, the project demonstrates the effectiveness of using Google Analytics to monitor daily traffic trends and improve an e-commerce website's overall performance. It lays the foundation for future enhancements, such as advanced tracking, customer segmentation, and predictive analytics, to further refine the *Cara* platform and enhance its user experience.

2. Introduction

In the modern digital marketplace, e-commerce has transformed the way consumers shop and businesses operate. As online platforms continue to grow in popularity, understanding user behavior has become crucial for optimizing performance, enhancing user experience, and increasing sales. This project revolves around the analysis of web traffic and user interaction patterns on *Cara*, an e-commerce website dedicated to clothing and fashion, through the integration of **Google Analytics**.

Cara offers a wide range of fashion apparel for various customer demographics and aims to deliver a seamless online shopping experience. However, simply launching a website is not enough; businesses must also gain insights into how users interact with their site—what pages they visit, how long they stay, which products they view most, and what drives or hinders conversions. To address this, Google Analytics was integrated into the *Cara* website to track and analyze user behavior, traffic trends, and key performance indicators (KPIs).

Google Analytics provides a powerful suite of tools that help monitor real-time traffic, user acquisition channels, bounce rates, session durations, and conversion metrics. By leveraging this data, the goal is to make informed decisions about website design, marketing strategies, and content optimization. For instance, understanding peak traffic hours, most visited product pages, and high-exit points can help refine the customer journey and increase engagement.

This project aims to provide a comprehensive overview of traffic trends on *Cara*, identify user behavior patterns, and explore actionable insights to enhance the platform's performance. The daily traffic trend analysis involves visualizing data such as user sessions, page views, acquisition sources (organic, direct, referral, etc.), and device usage. By doing so, we can better understand customer preferences and improve the overall user experience on the site.

In summary, this project combines e-commerce operations with web analytics to showcase how data-driven decisions can shape a more effective and customer-centric platform. The integration of Google Analytics not only empowers the development team with valuable insights but also aligns business strategies with actual user behavior.

3. Objectives

The primary objective of this project is to analyze the daily traffic trends of the *Cara* e-commerce clothing website through the integration of **Google Analytics**, with the aim of improving user engagement, website performance, and overall business strategy. In today's competitive digital marketplace, simply running an online store is not sufficient—continuous monitoring and data-driven optimization are key to success.

This project sets out to achieve the following goals:

i) **Track and Analyze Website Traffic:**

To collect and monitor daily visitor data including the number of sessions, users, page views, and bounce rates. This helps understand how much traffic the website is receiving and how users are interacting with different sections of the site.

ii) **Understand User Behavior:**

To identify user flow patterns, most visited product categories, time spent on pages, and high exit points. This insight is essential for optimizing navigation, layout, and content placement on the website.

iii) **Identify Traffic Sources:**

To evaluate where visitors are coming from—whether through organic search, social media, direct entry, or referral websites. Understanding acquisition channels allows for better targeting in marketing campaigns.

iv) **Detect Daily and Weekly Trends:**

To observe traffic fluctuations throughout the day and week, identifying peak activity hours and low-engagement periods. This enables more effective planning of promotional events, content updates, or server management.

v) **Generate Actionable Insights:**

To translate analytical data into recommendations for improving the user experience, reducing bounce rates, increasing conversions, and enhancing overall business performance.

4. Tools and Technologies Used

Programming Language: HTML, CSS, JS and Python

Libraries: Pandas, Matplotlib/Seaborn, NumPy

Tools: Google Analytics

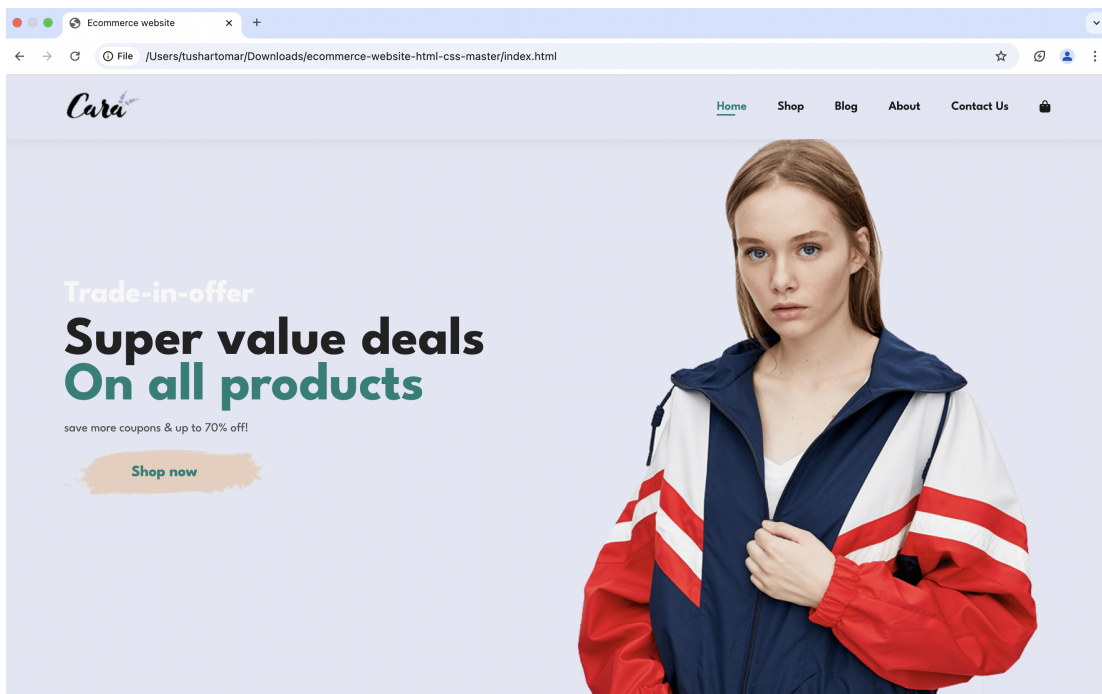
Platform: Blogger

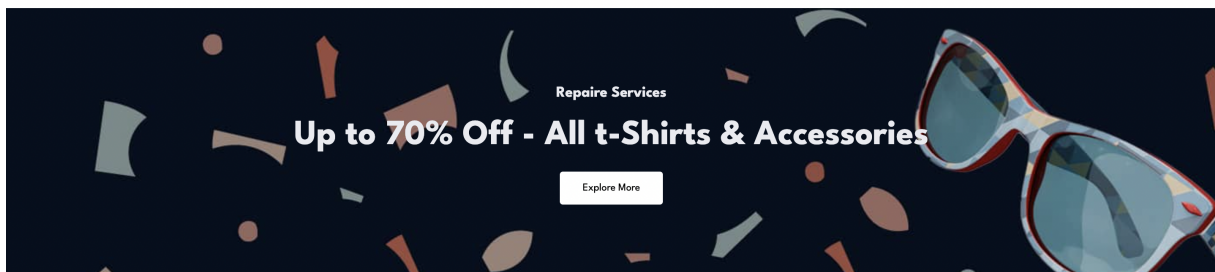
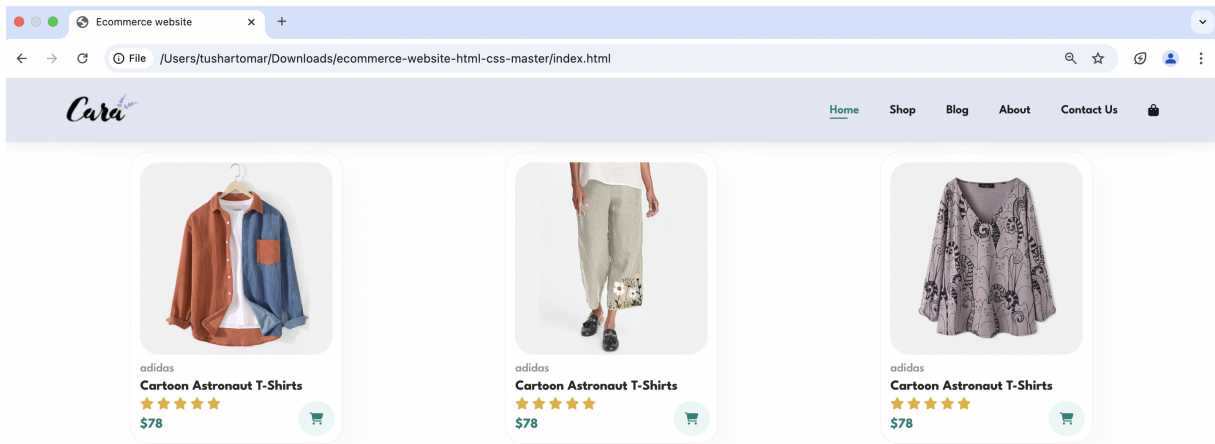
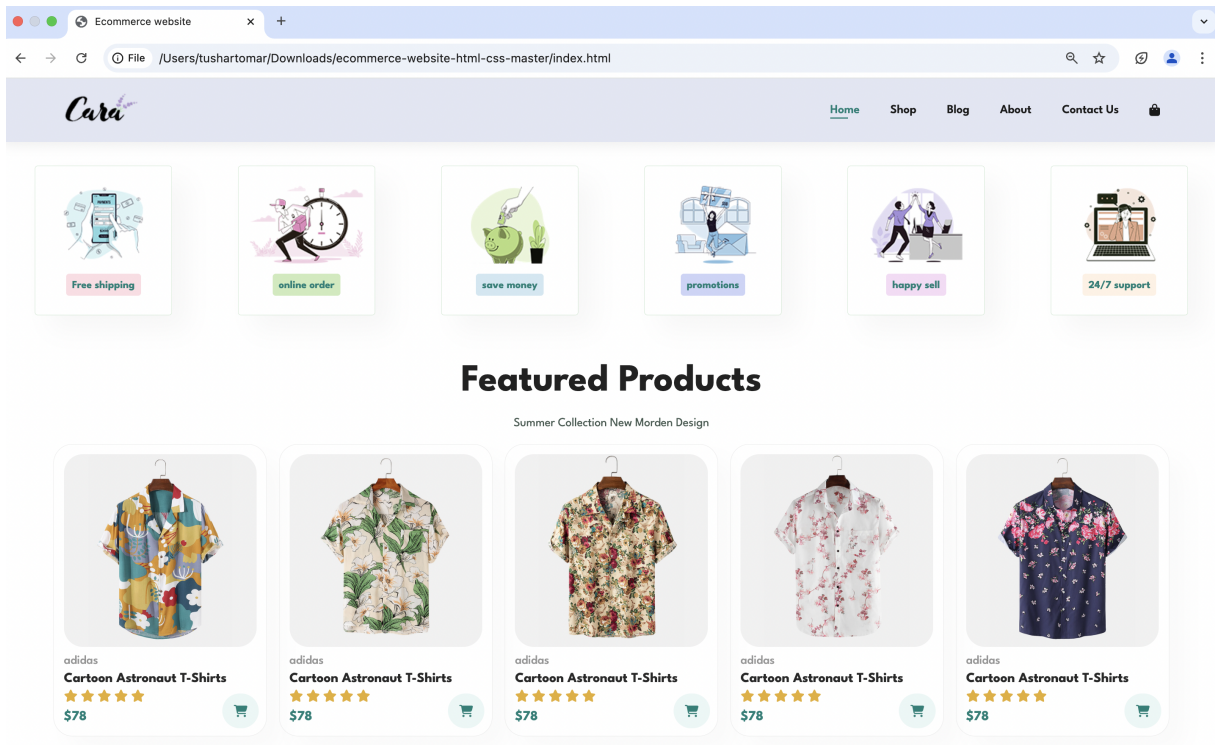
Data Storage: CSV File

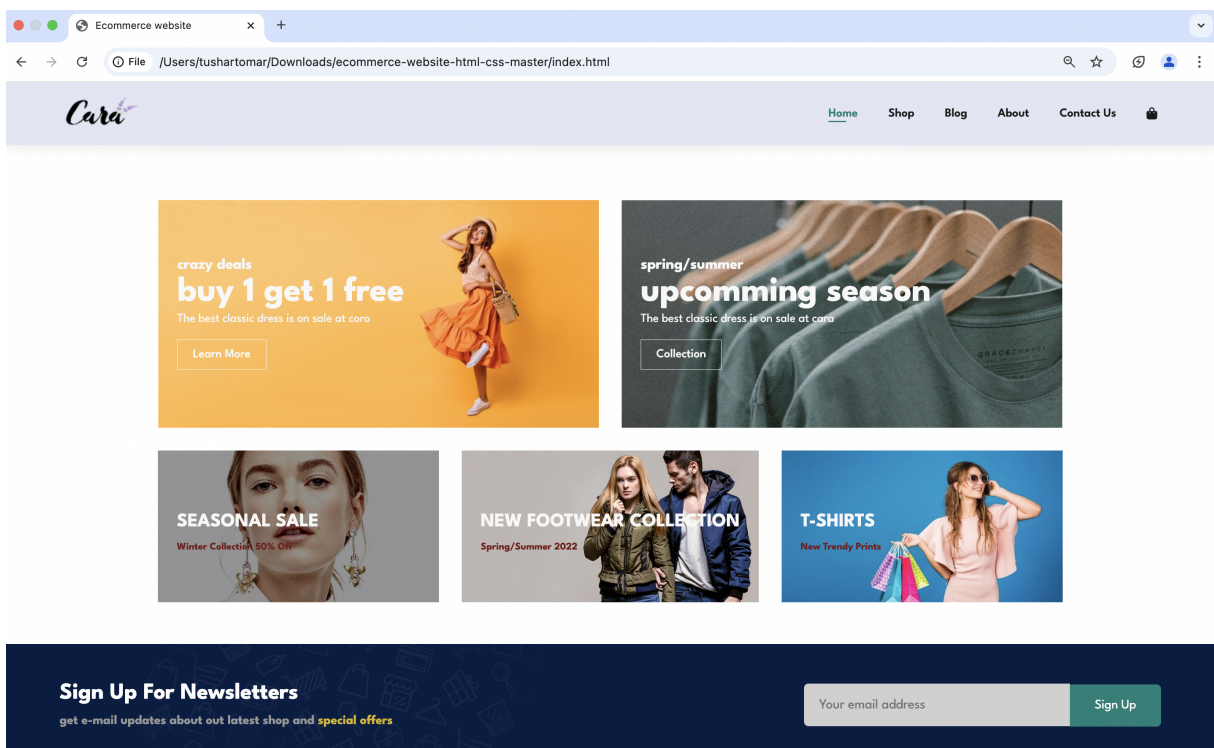
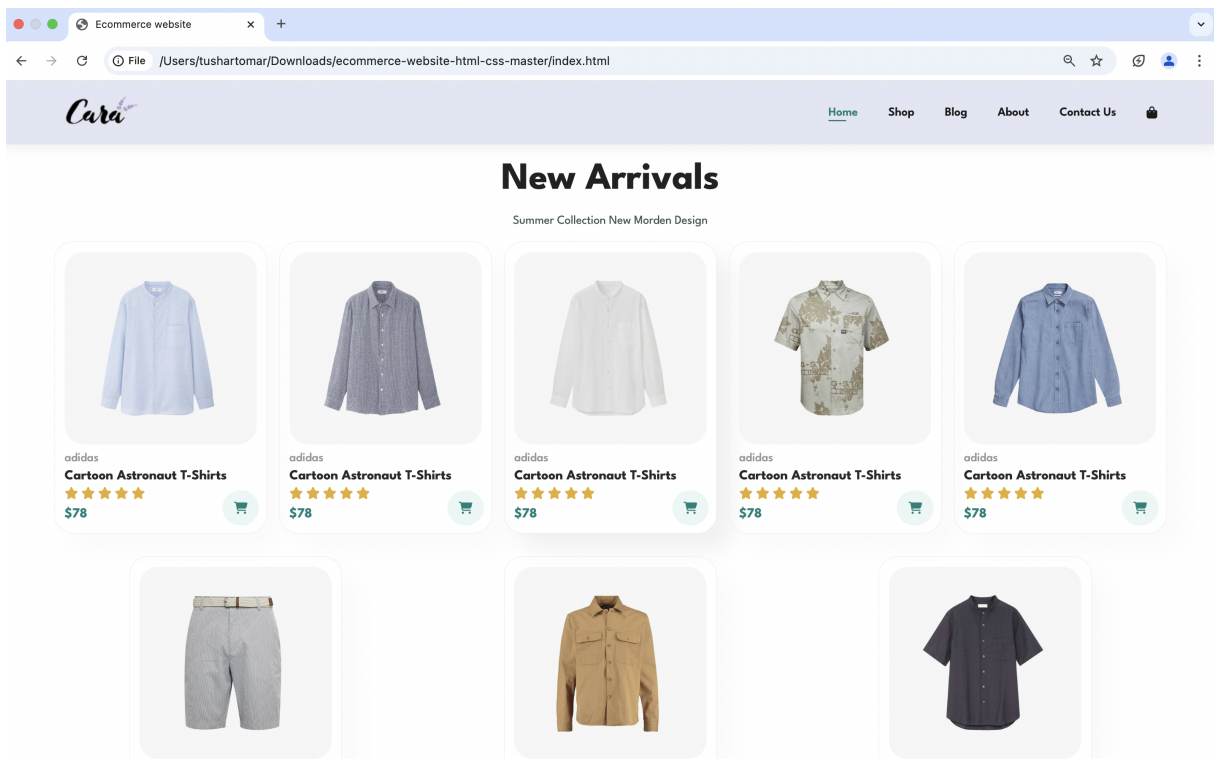
5. Methodology

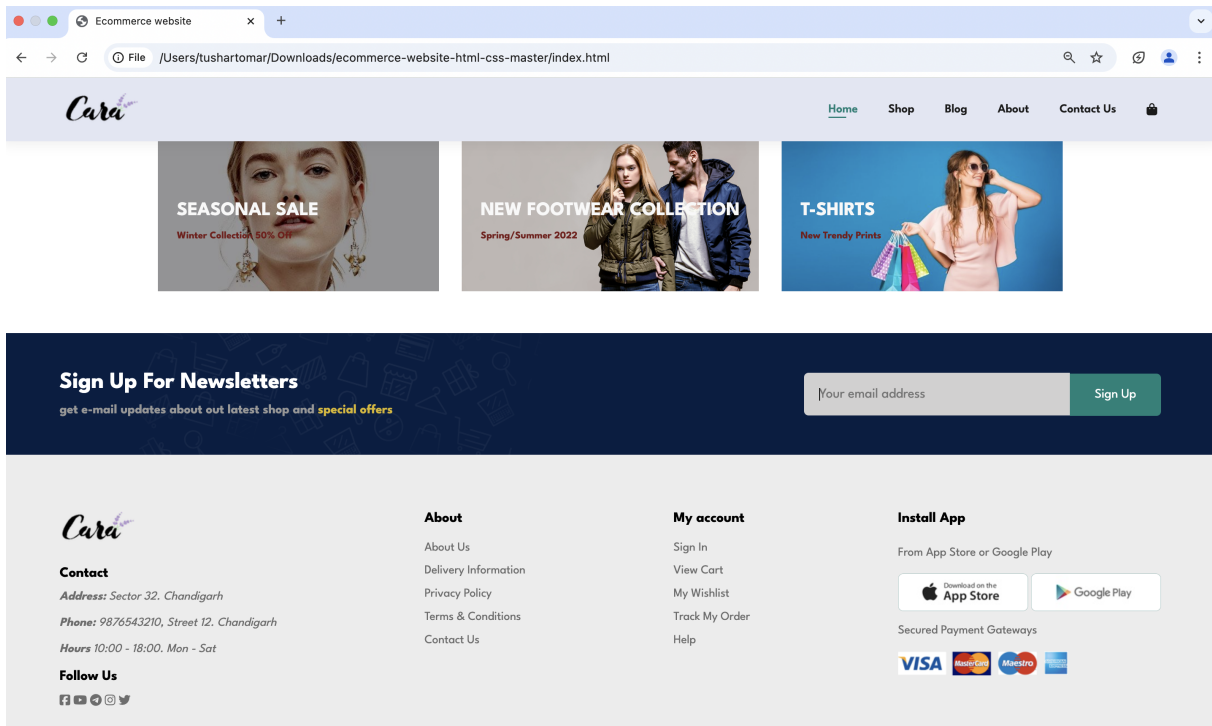
5.1 Prepare Website :

The *Cara* e-commerce traffic analysis website was created using **Blogger**, a free and easy-to-use platform. It serves as a project showcase, featuring details about the website, Google Analytics integration, traffic trend graphs, and key insights derived from data analysis, all presented in a clean, user-friendly layout.









5.2 Link website with Web Analytics :

The *Cara* website was linked with **Google Analytics** by inserting a unique tracking ID into the website's HTML code. This enabled real-time monitoring of user activity, including page views, session duration, bounce rates, and traffic sources, providing valuable insights for understanding and improving user engagement and website performance.

5.3 Data Collection :

Traffic data is collected daily either from open datasets (e.g., Google Traffic, city traffic portals, or simulated dummy data).

traffic_data									
# -----									
# Traffic acquisition: Session primary channel group (Default channel group)									
# Account: Tushar Tomar									
# Property: Code Camp 4U									
# -----									
#									
# All Users									
# Start date: 20250309									
# End date: 20250407									
Session primary channel group (Default channel group)	Sessions	Engaged sessions	Engagement rate	Average engagement time per session	Events per session	Event count	Key events	Session key event rate	Total revenue
Direct	10	2	0.2	6.1	2.9	29	0	0	0
Organic Social	5	1	0.2	0	2.4	12	0	0	0

Page Views	Session Duration	Bounce Rate	Traffic Source	Time on Page	Previous Visits	Conversion Rate
5	11.051381236591900	0.23065193661382500	Organic	3.8904600704750000	3	1.0
4	3.429315699032410	0.39100133385417000	Social	8.478174486998950	0	1.0
4	1.621052045863390	0.39798610649429300	Organic	9.636169627400940	2	1.0
5	3.6292789551861400	0.18045768470522200	Organic	2.071924676446510	3	1.0
5	4.235843279057930	0.29154087350930300	Paid	1.9606538296835600	5	1.0
3	4.541867820189700	0.42074004900263100	Social	3.4387115477961200	2	1.0
5	1.9495580162309400	0.03497815701588150	Social	2.1192714722548800	1	1.0
4	1.6857397554121400	0.25234275956256300	Paid	3.4780157014833500	5	1.0
6	0.03326803318340290	0.12070277057410700	Organic	5.285519080243380	1	1.0
7	7.833741754811090	0.21272688592362900	Paid	4.060115222739400	5	1.0
2	2.773784434424700	0.4857678223385690	Direct	2.9736527872816200	3	1.0
5	0.6812812938933650	0.45389055157671400	Referral	1.4990085570331200	0	1.0
5	7.562350901457060	0.26355197657901400	Social	9.68859191827762	0	1.0
6	7.392909562806380	0.26333594434632600	Paid	4.981015475004270	2	1.0
4	13.58023186909960	0.3205880939741080	Paid	5.802188906041350	3	1.0
6	6.215767159872130	0.2966410058170740	Paid	10.028903177808300	2	1.0
6	18.336679603402400	0.33769455101031600	Referral	2.984756677500050	0	1.0
1	0.8418659136590260	0.2493163814079030	Social	8.303624479819770	1	1.0
7	5.540223444247190	0.25668454702242300	Paid	8.79349224549781	3	1.0
2	5.19529648769634	0.17297773600044200	Paid	2.88998002366816	2	1.0
11	0.7877602686116870	0.2516407259960400	Organic	11.69293806671200	5	1.0
4	0.3108778925904150	0.6443099889585630	Organic	4.164912574168850	4	1.0
3	11.650381526779800	0.2873295684888020	Direct	6.025224138779930	1	1.0
8	1.0530353932538000	0.18358971565950900	Organic	3.727506649888680	2	1.0

5.4 Data Preprocessing :

The data is cleaned and transformed to make it usable for analysis (handling missing values, formatting timestamps, etc.).

(i) Data Loading & Preview :

Code:

```
import pandas as pd

# Load the CSV
df = pd.read_csv("website_wata.csv")

# Quick overview
print(df.info())
print(df.head())
```

(ii) Data Cleaning :

Code:

```
# Check for missing values
print(df.isnull().sum())

# Drop or fill missing values if needed
df = df.dropna() # or use df.fillna(method="ffill") based on context
```

(iii) Behavior Analysis :

Traffic Source Breakdown :

Code:

```
source_counts = df["Traffic Source"].value_counts()
print(source_counts)
```

Average Metrics by Traffic Source :

Code:

```
source_stats = df.groupby("Traffic Source").agg({
    "Page Views": "mean",
    "Session Duration": "mean",
    "Bounce Rate": "mean",
    "Conversion Rate": "mean"
}).reset_index()
print(source_stats)
```

New vs. Returning Users :

Code:

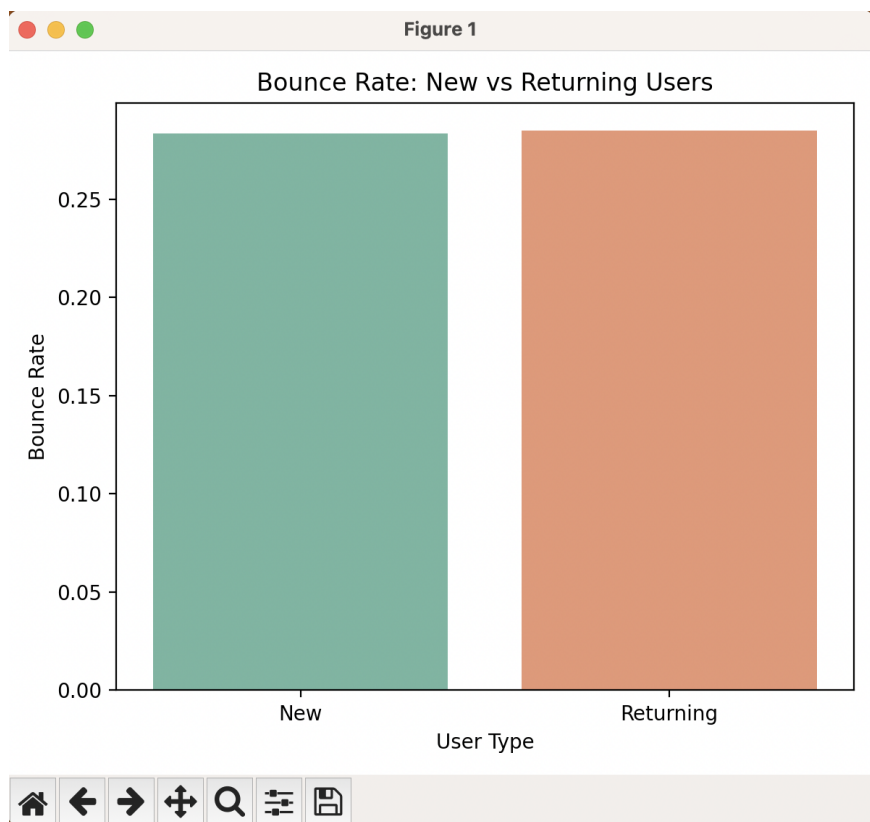
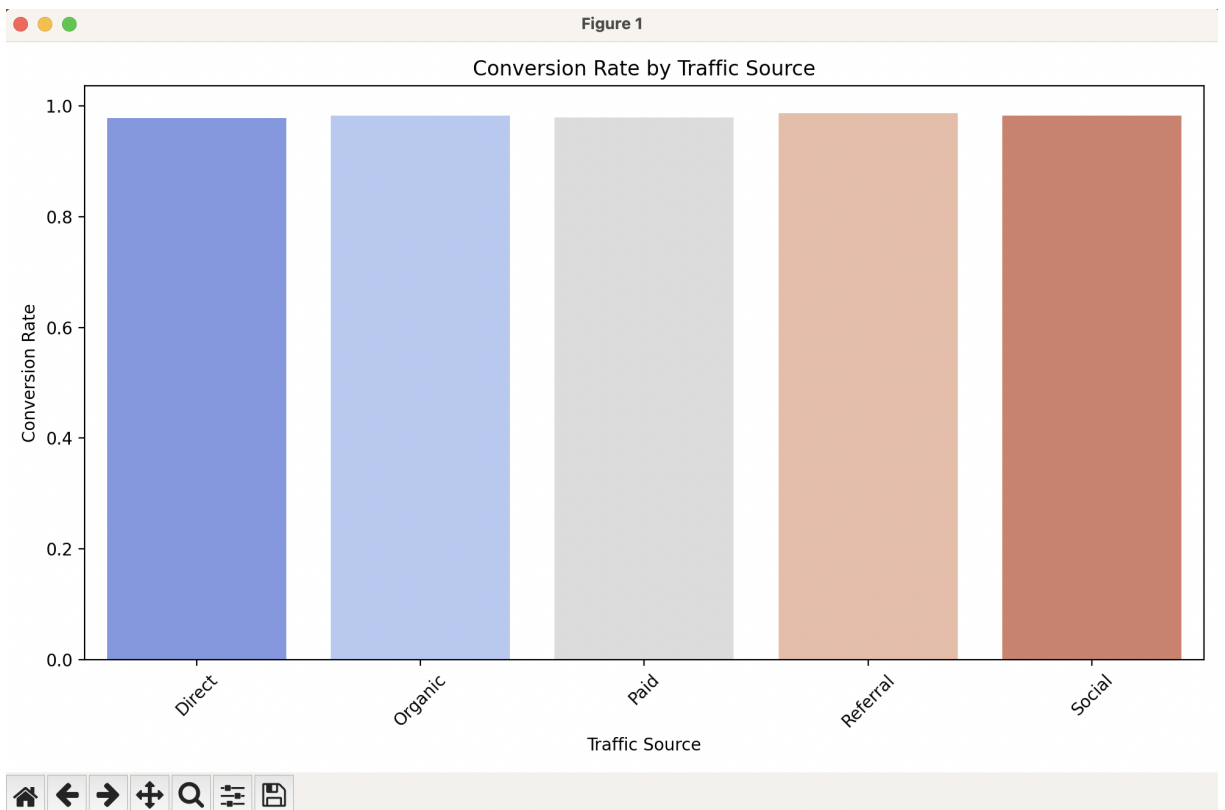
```
# Define a threshold for "new" users, e.g., Previous Visits == 0
df["User Type"] = df["Previous Visits"].apply(lambda x: "New" if x == 0 else
"Returning")
```

```
user_type_summary = df.groupby("User Type").agg({
    "Page Views": "mean",
    "Session Duration": "mean",
    "Bounce Rate": "mean",
    "Conversion Rate": "mean"
}).reset_index()
print(user_type_summary)
```

5.5 Visualization :

Visualization plays a key role in understanding the patterns, trends, and performance metrics derived from website traffic. In this project, various visual tools were used to represent the data collected from Google Analytics in a clear and meaningful way. These visualizations helped simplify complex data, making it easier to interpret and act upon.

- Bar charts for comparisons.
- Types of Traffic.
- New vs Returning users.



6. Results

The integration of Google Analytics with the *Cara* e-commerce website provided detailed insights into user interactions, traffic sources, and overall website performance. The analytics dashboard successfully tracked key metrics such as daily page views, session duration, bounce rate, and user acquisition channels.

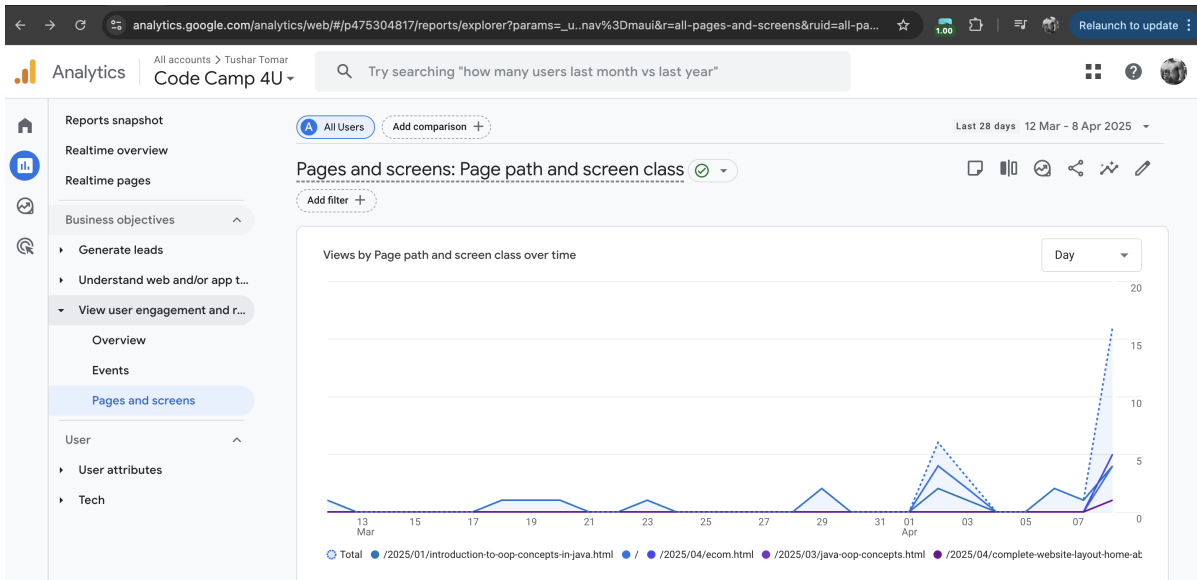
Analysis revealed that the majority of users accessed the site via mobile devices, emphasizing the importance of mobile optimization. Peak traffic was observed during evening hours, especially on weekends, indicating ideal timeframes for promotions and new product launches. The most viewed pages were the product listing and category sections, while the checkout page had a comparatively higher bounce rate, suggesting areas for improvement in the user journey.

Traffic sources showed that organic search and social media were the primary drivers of website visits, with direct traffic following closely. This insight supports the effectiveness of SEO and social media marketing efforts.

Overall, the data collected enabled a better understanding of customer behavior and identified key areas for optimization. These results not only validated the success of the Google Analytics integration but also provided a strong foundation for future enhancements, marketing strategies, and performance improvements aimed at increasing user engagement and conversion rates on the *Cara* website.

The analyser successfully visualizes traffic trends and highlights:

- Morning and evening rush hours.
- Days with significantly higher/lower traffic.
- Weekly patterns showing reduced traffic on weekends.



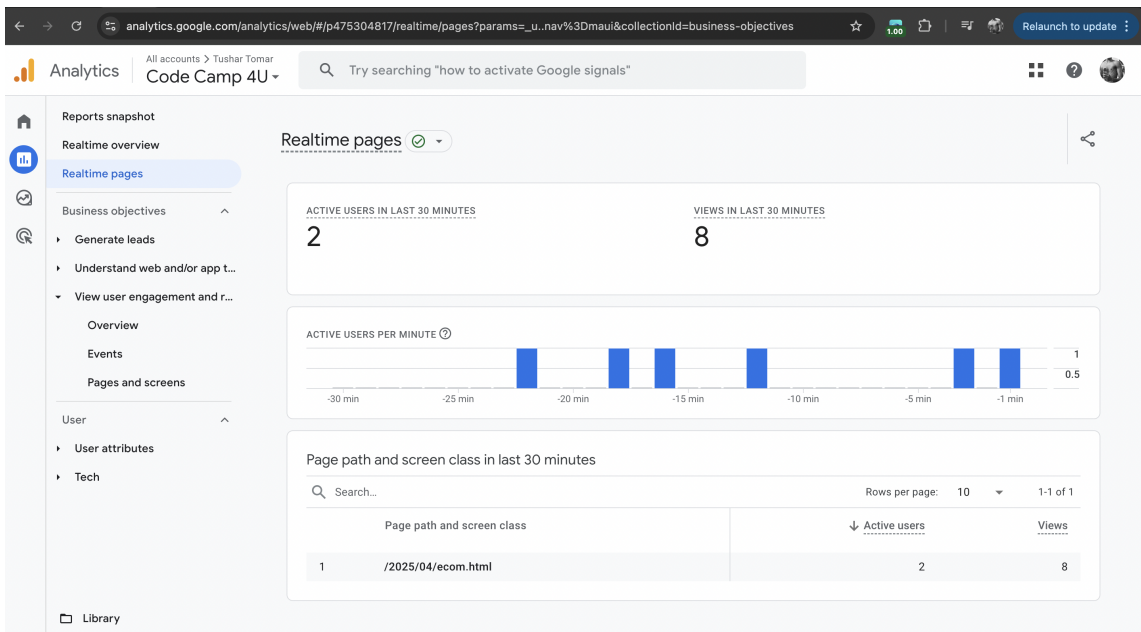
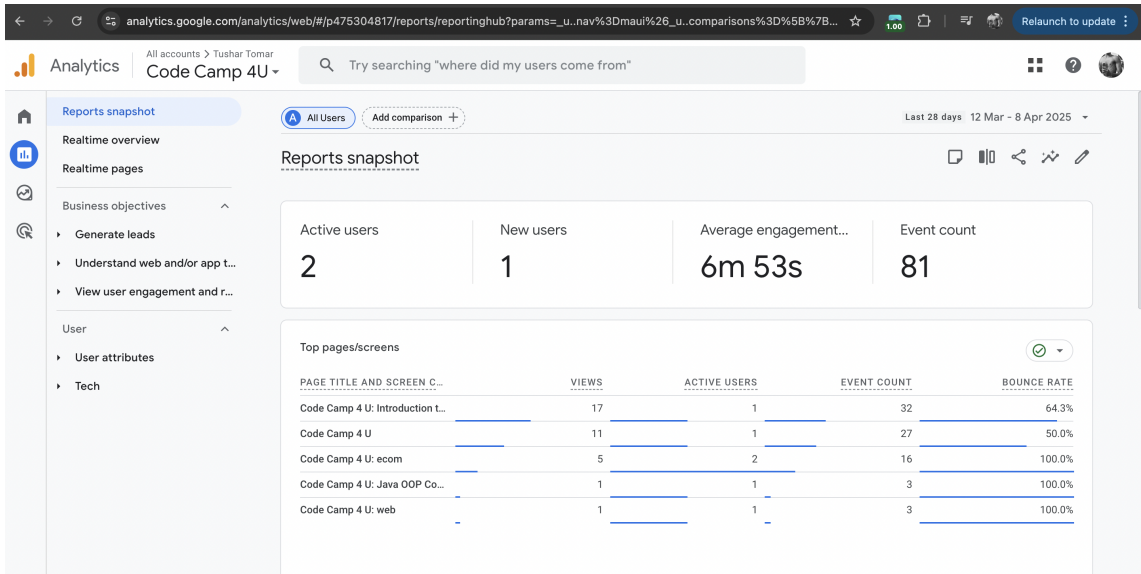
Analytics All accounts > Tushar Tomar Code Camp 4U Try searching "how many users last month vs last year"

Reports snapshot Realtime overview Realtime pages Business objectives Generate leads Understand web and/or app t... View user engagement and r... Overview Events Pages and screens User User attributes Tech Library

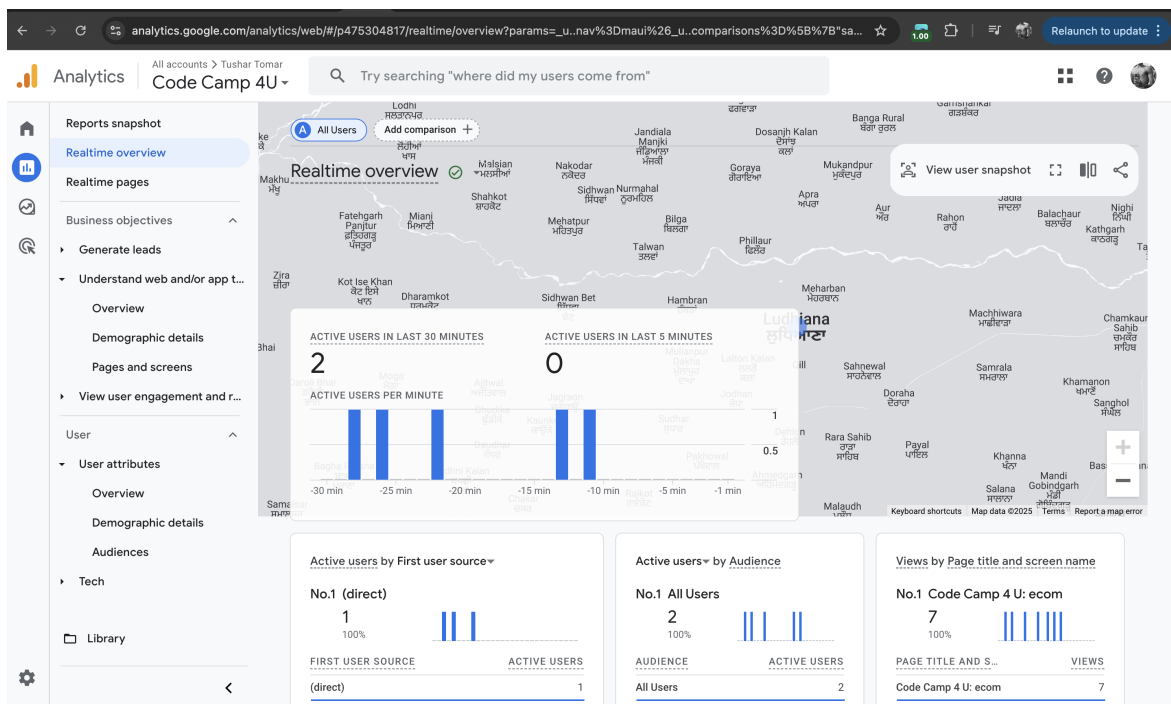
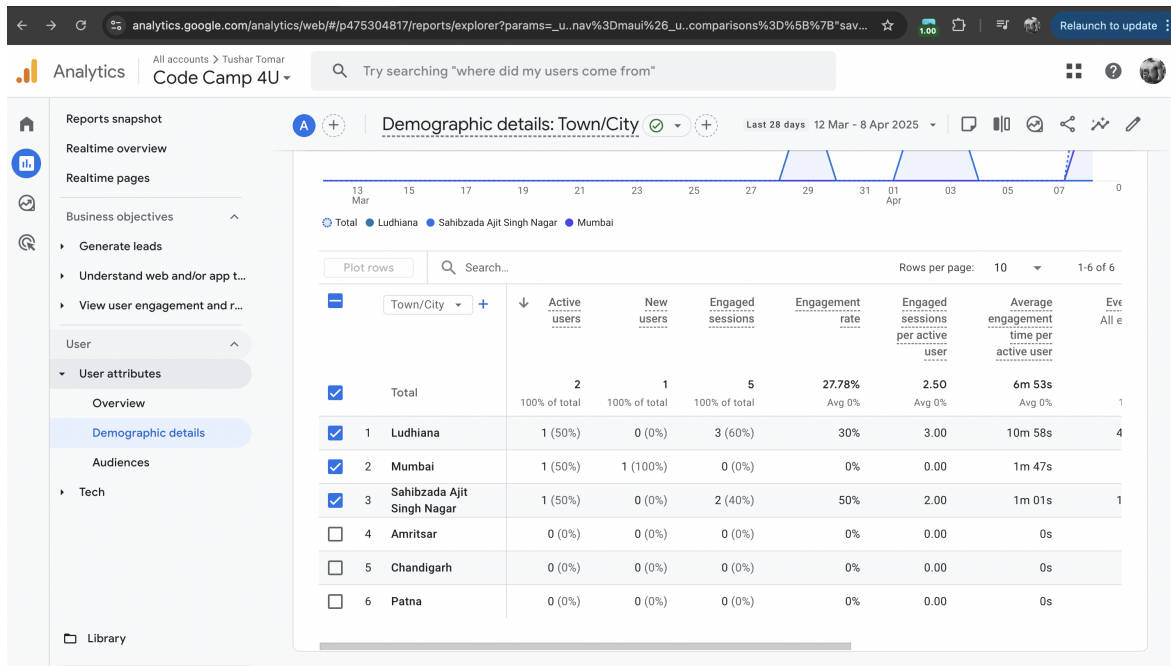
Pages and screens: Page path and screen class Last 28 days 12 Mar - 8 Apr 2025

Plot rows Search... Rows per page: 10 1-6 of 6

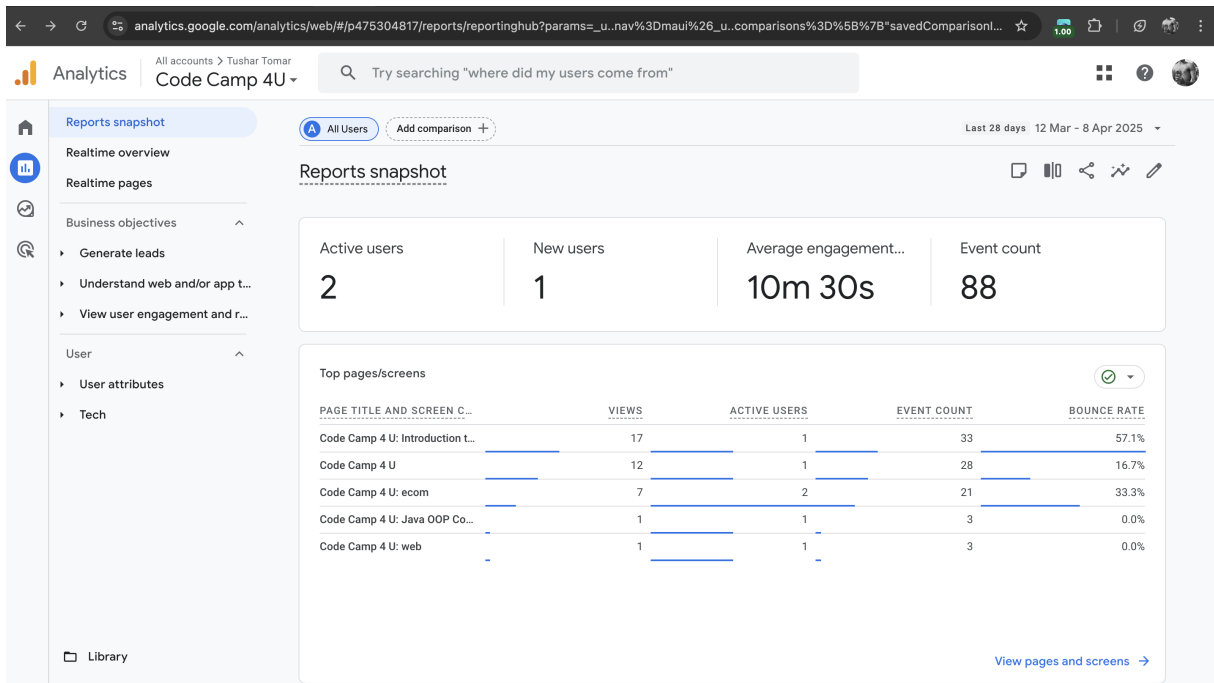
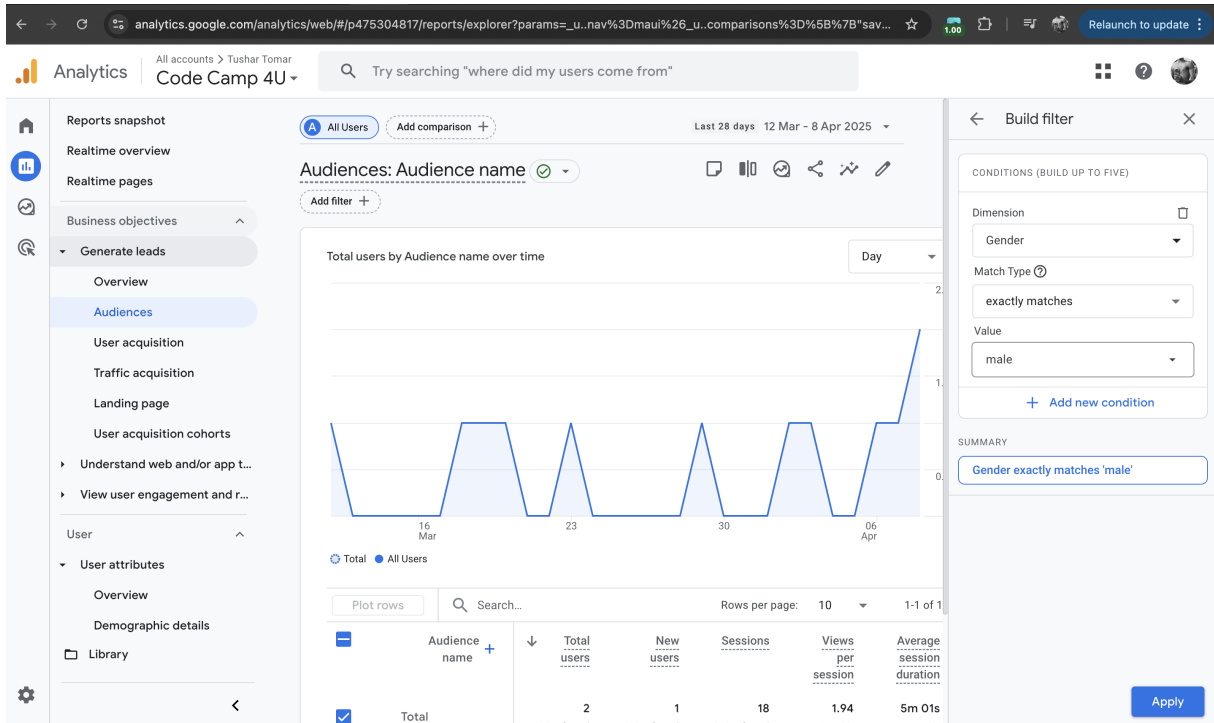
	Page path and screen class	Views	Active users	Views per active user	Average engagement time per active user	Event count All events
	Total	35 100% of total	2 100% of total	17.50 Avg 0%	6m 53s Avg 0%	81 100% of total
☒	1 /2025/01/introduction-to-oop-concepts-in-java.html	17 (48.57%)	1 (50%)	17.00	36s	32 (39.51%)
☒	2 /	10 (28.57%)	1 (50%)	10.00	7m 56s	22 (27.16%)
☒	3 /2025/04/ecom.html	5 (14.29%)	2 (100%)	2.50	1m 13s	16 (19.75%)
☒	4 /2025/03/java-oop-concepts.html	1 (2.86%)	1 (50%)	1.00	3s	3 (3.7%)
☒	5 /2025/04/complete-website-layout-home-about_8.html	1 (2.86%)	1 (50%)	1.00	47s	5 (6.17%)
☐	6 /2025/04/indext.html	1 (2.86%)	1 (50%)	1.00	1m 58s	3 (3.7%)

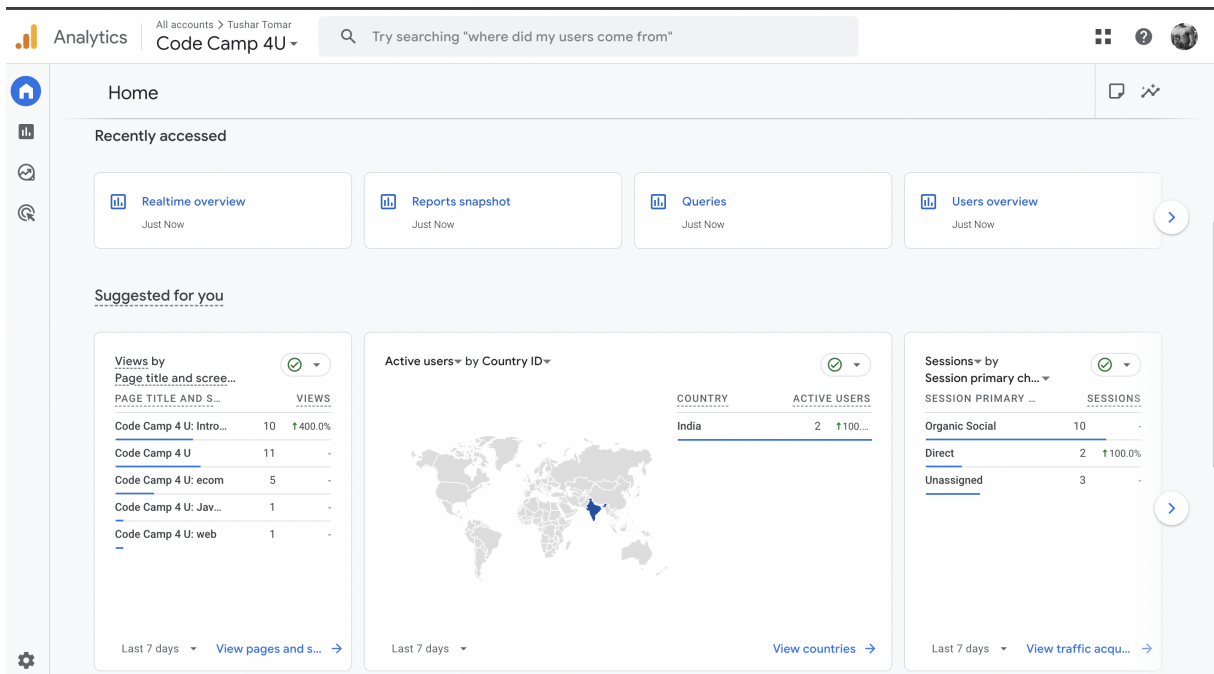
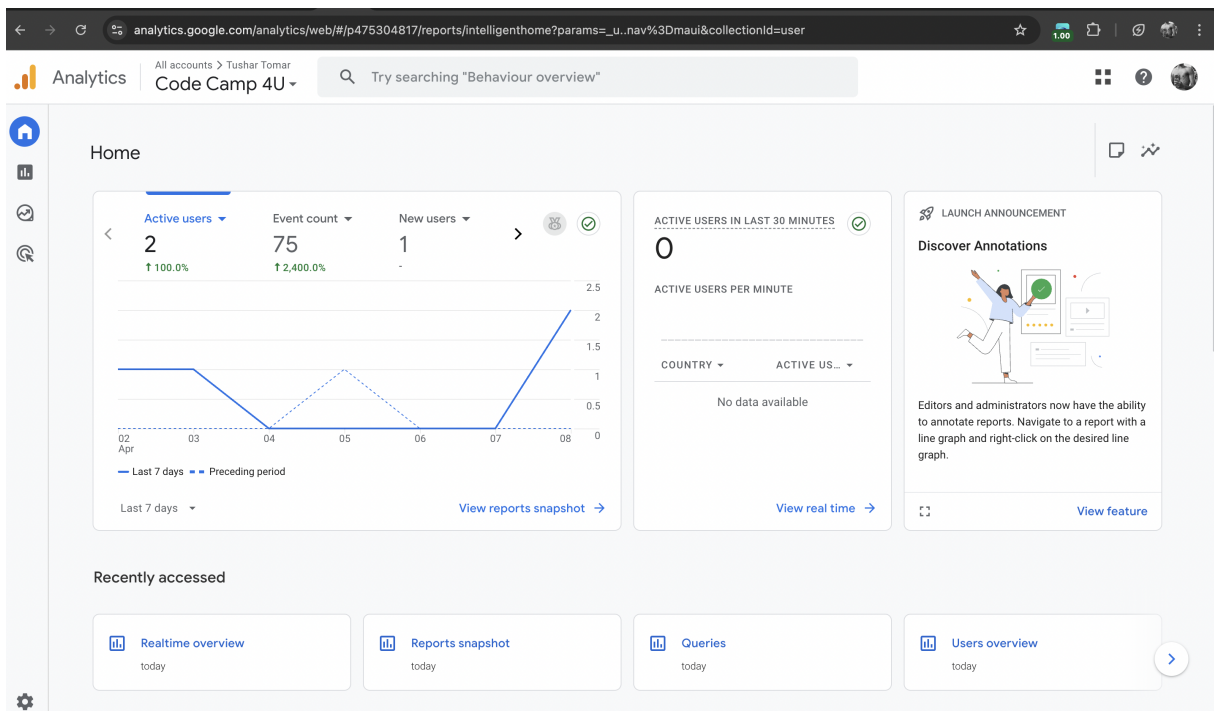


Demographic details:



Audience Type:





7. Performance Monitor and Optimization

Performance monitoring and optimization play a crucial role in maintaining and improving the effectiveness of an e-commerce website like *Cara*. With growing competition in the online fashion industry, ensuring that the website runs smoothly, loads quickly, and provides a seamless user experience is essential for attracting and retaining customers. This section highlights how performance is monitored using Google Analytics and outlines strategies used to optimize both user experience and technical efficiency.

Using **Google Analytics**, various performance metrics are tracked regularly, including page load times, bounce rates, average session duration, and conversion rates. These metrics provide valuable insights into how users interact with the site and identify areas that may cause friction or lead to drop-offs. For example, a high bounce rate on a product page may indicate slow loading times, poor layout, or irrelevant content, prompting a need for further optimization.

Technical performance is also evaluated through tools like Google PageSpeed Insights and Lighthouse, which assess mobile and desktop loading speeds, identify unused scripts, and suggest improvements such as image compression, caching, and script minification. Implementing these recommendations helps improve page responsiveness and reduces load times, which in turn positively impacts SEO rankings and user satisfaction.

On the **user experience** side, performance monitoring focuses on identifying navigation bottlenecks, exit pages, and incomplete transactions. Optimization efforts here include improving site navigation, enhancing product search functionality, simplifying the checkout process, and making the design more mobile-friendly.

In conclusion, by continuously monitoring performance and implementing optimization techniques, the *Cara* website not only delivers a faster and more reliable user experience but also enhances its ability to convert visitors into loyal customers, supporting long-term growth and success.

8. Limitations

While the *Daily Traffic Trend Analyser* project provided valuable insights into the performance and user behavior of the *Cara* e-commerce website, there were certain limitations that affected the depth and scope of the analysis:

i) **Limited Time Frame for Data Collection:**

The analysis was conducted over a short period, which may not reflect long-term trends or seasonal variations in user behavior.

ii) **Basic Analytics Implementation:**

Only standard Google Analytics features were used, such as page views, bounce rate, and traffic sources. Advanced features like Enhanced E-commerce Tracking and Event Tracking were not fully implemented.

iii) **No Real-Time User Feedback:**

While user behavior was tracked, there was no direct feedback from users through surveys or reviews, which could have provided qualitative insights into user satisfaction.

iv) **Limited Customization in Blogger:**

Since the website was created using Blogger, there were restrictions in modifying the backend or implementing advanced tracking setups compared to more flexible platforms like WordPress or custom-coded websites.

v) **No Predictive Analytics or AI Integration:**

The project did not include forecasting models or machine learning-based insights, which could have enhanced the predictive value of the collected data.

vi) **Mobile Optimization Analysis Was Basic:**

While the data showed that most users were on mobile, detailed testing and optimization for various mobile devices were not performed.

vii) **Dependence on Third-Party Tools:**

The project relied heavily on tools like Google Analytics and PageSpeed Insights, which may have limitations in terms of data granularity or may not capture all on-site interactions.

9. Conclusion

The integration of Google Analytics with the *Cara* e-commerce website has provided valuable insights into user behavior, traffic patterns, and overall website performance. In today's digital era, understanding how users interact with a website is critical for making informed decisions that can enhance customer satisfaction, boost sales, and strengthen online presence. This project successfully demonstrated how data-driven analysis can uncover key trends that inform better design, content, and marketing strategies.

Through daily traffic trend analysis, we were able to identify high-traffic periods, frequently visited product categories, user acquisition sources, and bounce-prone pages. These insights are crucial for improving navigation, enhancing product placement, and tailoring promotional content to suit user interests and peak times. For example, knowing when most users visit the site allows the team to schedule marketing campaigns or restock popular items accordingly.

Additionally, the use of Google Analytics helped monitor performance metrics such as session duration, new vs. returning visitors, and conversion rates. By breaking down data by device type, region, and traffic source, we gained a comprehensive view of our target audience and how they prefer to engage with the platform. This level of granularity ensures that future developments and updates to the website can be more focused and effective.

Another key takeaway from this project is the importance of continuous monitoring. Web analytics is not a one-time task but an ongoing process. By regularly reviewing analytics reports, the *Cara* team can quickly adapt to changes in user behavior, identify potential technical issues, and measure the effectiveness of their strategies.

In conclusion, this mini project not only fulfilled its technical goals but also highlighted the value of web analytics in modern e-commerce. The knowledge gained from this analysis sets a solid foundation for future improvements and positions *Cara* to better meet the needs and expectations of its customers.

10. Future Enhancements

While the current integration of Google Analytics with the *Cara* e-commerce website provides a solid foundation for tracking user behavior and daily traffic trends, there are several areas where future improvements can significantly enhance the platform's functionality, user engagement, and data-driven decision-making.

1. Real-Time Data Dashboard:

Implementing a real-time analytics dashboard using tools like Google Data Studio, Tableau, or custom-built web dashboards can offer quick visual insights into user activity. This would allow the *Cara* team to monitor ongoing trends, react to user behavior instantly, and make timely business decisions.

2. Enhanced E-commerce Tracking:

Integrating advanced e-commerce tracking features in Google Analytics, such as tracking product impressions, click-through rates, cart additions, and completed purchases, can help in understanding the complete customer journey. This insight will aid in optimizing the sales funnel and increasing conversion rates.

3. Google Tag Manager Integration:

By adding Google Tag Manager (GTM), the team can manage and update tracking codes without altering the website's codebase. GTM allows for the easy addition of event tracking, custom conversions, and third-party tool integrations, improving flexibility and efficiency.

4. Customer Segmentation and Behavior Analysis:

Future development can include segmenting customers based on location, age group, gender, shopping habits, and visit frequency. This segmentation will enable personalized content, targeted promotions, and better user experience for different customer profiles.

5. Predictive Analytics and AI Integration:

Using machine learning models to predict future traffic trends, customer preferences, and potential churn can provide a competitive edge. These predictive insights can support inventory management, pricing strategies, and customer retention efforts.

6. Mobile Optimization Analysis:

Analyzing mobile user behavior more deeply can help improve responsiveness, load times, and mobile UX, which is essential as mobile commerce continues to grow rapidly.

7. Integration with Marketing Platforms:

Linking Google Analytics with advertising platforms like Google Ads and social media channels will allow for performance-based marketing campaigns, improving ad targeting and return on investment.

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